

115. What is the value of n ?

- (a) 4 (b) 5 (c) 6 (d) 7

116. What is the value of $p + q$?

- (a) $\frac{57}{32}$ (b) $\frac{53}{32}$ (c) $\frac{51}{32}$ (d) 1

Consider the following for the next **two (02)** items that follow:

x_i	1	2	3	...	n
y_i	1	2^{-1}	2^{-2}	...	$2^{-(n-1)}$

117. What is $\sum_i^n x_i f_i$ equal to?

- (a) $\frac{2^{n+1} - n + 2}{2^{n-1}}$ (b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$
 (c) $\frac{2^{n+1} + n + 2}{2^{n-1}}$ (d) $\frac{2^{n+1} - n - 2}{2^n}$

118. What is the mean of the distribution?

- (a) $\frac{2^{n+1} - n + 2}{2^n - 1}$ (b) $\frac{2^{n+1} - n - 2}{2^{n-1}}$
 (c) $\frac{2^{n+1} - n - 2}{2^n - 1}$ (d) $\frac{2^{n+1} - n + 2}{2^n}$

Consider the following for the next **two (02)** items that follow:

The marks obtained by 10 students in a Statistics test are 24, 47, 18, 32, 19, 15, 21, 35, 50 and 41.

119. What is the mean deviation of the largest five observations?

- (a) 4.8 (b) 5.5 (c) 6 (d) 7.5

120. What is the variance of the largest five observations?

- (a) 14.6 (b) 21.8 (c) 25.2 (d) 46.8

Answer Key

Q. No	Answer Key	Topic's Name	Chapter's Name
1	d	Adjoint Matrix	Matrices and Determinants
2	b	Power of Complex Number	Complex Numbers
3	d	Sum of Number	Permutations and Combinations
4	a	Cube roots of Units	Complex Numbers
5	c	Combination	Permutations and Combinations
6	a	Combination	Permutations and Combinations
7	a	Logarithm	Relations and Functions
8	c	Roots of equation	Theory of equations
9	c	Modulus	Complex Numbers
10	b	Power of Prime Number	Permutations and Combinations
11	b	Adjoint Matrix	Matrices and Determinants
12	d	Identity matrix	Matrices and Determinants
13	b	Permutation	Permutations and Combinations
14	b	Cube roots of Units	Complex Numbers
15	d	Power of complex number	Complex Numbers
16	a	Skew Symmetric Matrix	Matrices and Determinants
17	d	Permutation	Permutation and Combination
18	a	Remainder	Binomial
19	c	Value of Determinant	Matrices and Determinants
20	b	Value of Determinant	Matrices and Determinants
21	d	Number of Matrices	Matrices and Determinants
22	a	One-One & Onto function	Relations and Functions
23	d	Equivalence Relation	Relations and Functions
24	a	Operation on Sets	Sets
25	a	Values of Determinant	Matrices and Determinants
26	a	A.P. and G.P.	Sequence and Series
27	b	Logarithmic Function	Relations and Functions